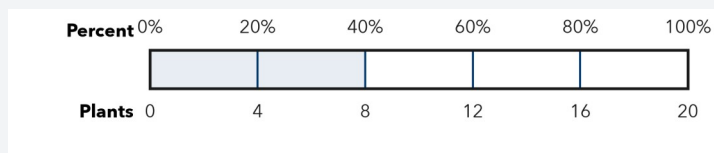


LESSON 4-1 • SESSION 1 • EXPLAIN YOUR THINKING**Percent Means Per Hundred**

Study the model. Then solve your own problem the same way. Use only the help you need.

WORDS percent (*por ciento*) • decimal grid (*cuadrícula decimal*) • equivalent ratio (*razón equivalente*)**1 • SEE A MODEL***read it with a partner***SEE A MODEL** *a problem like yours, solved*

A garden has 20 plants. 8 of them are tomato plants. What percent of the plants are tomatoes?

**HOW I SOLVED IT**

- 1 Name the whole.** All 20 plants = 100%.
- 2 Find one part.** 5 equal parts: $20 \div 5 = 4$ plants, and $100\% \div 5 = 20\%$.
- 3 Count parts.** 8 plants fill 2 parts. $2 \times 20\% = 40\%$.

I SAY

"The whole is **20** plants, so each part is **20%**. 8 plants fill **2** parts, so **40%** are tomatoes."

I WRITE

I know the whole is **20 plants** because **the garden has 20 plants in all**. Each part of the bar is **20%**, so **8 plants is $2 \times 20\% = 40\%$** .

2 • DRAW IT*your turn: label the whole, the part and the percent*

A team finished 9 of 12 challenges at field day. Use the bar to show what percent of the challenges they finished.

**TRY THESE STEPS** *fill in the blanks as you work*

- 1** Name the whole: _____ challenges = 100%.
- 2** Find one part: $12 \div 4 =$ _____ challenges = _____%.
- 3** Count the parts that 9 challenges fill.

3 • SAY IT*tell a partner*

Partner A: "The whole is _____ challenges, so each part is _____%." Partner B: "9 challenges fill _____ parts, so the team finished _____%."

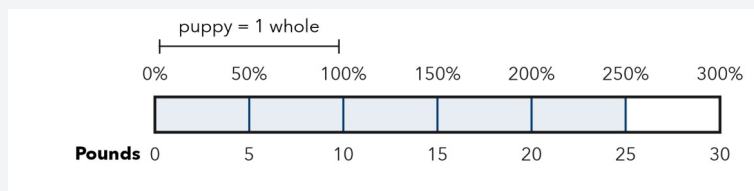
4 • WRITE IT*pick a starter • name the whole • use units***WORD BANK** the whole • each part • equal parts • fill • out of • percent**Because** I know the whole is _____ because _____.**But** Some students think 9 of 12 is 9%, but _____.**So** Each part of the bar is _____%, so _____.

LESSON 4-1 • SESSION 2 • EXPLAIN YOUR THINKING**Percents Greater Than 100%**

Study the model. Then solve your own problem the same way. Use only the help you need.

WORDS the whole (*el entero*) • times as much (*veces tanto como*) • greater than 100% (*mayor que 100%*)**1 • SEE A MODEL***read it with a partner***SEE A MODEL** *a problem like yours, solved*

A puppy weighs 10 pounds. Its mother weighs 25 pounds. The mother's weight is what percent of the puppy's weight?

**HOW I SOLVED IT**

- 1 Name the whole.** The puppy, 10 pounds = 100%.
- 2 Find one part.** 10 pounds is 2 parts, so 1 part = 5 pounds = 50%.
- 3 Count parts.** 25 pounds fills 5 parts: $5 \times 50\% = 250\%$.

I SAY

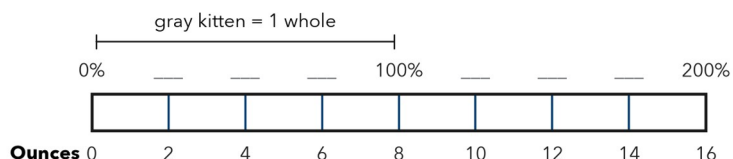
"The whole is **10** pounds, so each part is **50%**. 25 pounds fills **5** parts, so it is **250%**, more than 100%."

I WRITE

I know **the puppy** is the whole because **we compare the mother to the puppy**. She weighs more than one whole, so **her weight is 250%**.

2 • DRAW IT*your turn: label the whole, the part and the percent*

A gray kitten weighs 8 ounces. Its brother weighs 14 ounces. Use the bar to show the brother's weight as a percent of the gray kitten's weight.

**TRY THESE STEPS** *fill in the blanks as you work*

- 1** Name the whole: the gray kitten, _____ ounces = 100%.
- 2** Find one part: $8 \div 4 =$ _____ ounces = _____%.
- 3** Count the parts that 14 ounces fills.

3 • SAY IT*tell a partner*

Partner A: "The whole is _____ ounces, so each part is _____%." Partner B: "14 ounces fills _____ parts, so it is _____%, more than 100%."

4 • WRITE IT*pick a starter • name the whole • use units***WORD BANK** the whole • compare to • more than 100% • one whole • times as much • each part**Because** I know _____ is the whole because _____.**But** Some students think a percent cannot be more than 100%, but _____.**So** The brother weighs more than one whole, so _____.